

Allen Sun

Mechanical design engineer, Pittsburgh, PA

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Experience

Curtiss-Wright, Electro-Mechanical Division

Mar 2026 - present

Mechanical Design Engineer, Pittsburgh, PA

I design mechanical subsystems for large electric motors, from early layout through design for manufacturing, and make the drawings that go with them (GD&T per ASME Y14.5). The other half of the job is nonconforming hardware: working out with FEA, hand calcs and inspection data whether a part can be reworked or used as-is, then writing up the disposition.

Ergami Endoscopy

Jan - May 2025

R&D / Robotics Engineering Intern, New York, NY

- Designed and built the weld rig behind the TPU tube process for a soft robotic colonoscope insert: machined aluminum and brass with 3D-printed parts, for consistent heat welds.
- Worked on the insert's locomotion concept: anchor design, actuator selection and motion testing.

Relavo Medical

Sep - Dec 2024

Mechanical Engineering Intern, New York, NY

- Worked on a device that disinfects peritoneal dialysis tube connections.
- Reviewed the design control traceability matrix, wrote verification and validation test protocols, and designed test fixtures in SolidWorks, including torque and robustness tests.
- Made GD&T drawings for injection-molded parts.

Terrament

Sep 2023 - Sep 2024

Mechanical Engineering Intern, New York, NY

- Built and wired the 1/24-scale geared drive prototype shown at Newlab's New Climate Futures in September 2023.
- Load calculations, gear-ratio analyses and structural feasibility studies in Fusion 360 for a gravity storage system and its linear harmonic drive redesign.
- 3D-printed prototypes with off-the-shelf industrial parts; FEA for stress distribution, fatigue life and material selection.

Projects

Micro-vibration canceller

Senior design, NYU, Sep 2024 - May 2025

Bolt-on active vibration canceller (voice coil motor, piezo sensor). Machined every part and the test rig; open-loop tests cut beam vibration 75 to 90% at 54 Hz. Best Senior Design Project, NYU ME class of 2025.

Robotic arm

Advanced CAD, NYU, Fall 2024

Designed the hand, palm, wrist and forearm of a human-scale arm in Fusion 360; wrist FEA at 1000 N in SolidWorks Simulation.

Education

NYU Tandon School of Engineering, B.S. Mechanical Engineering. GPA 3.6/4.0

2025

Skills

Design

Mechanical and product design, DFM, GD&T (ASME Y14.5), fixture design, rapid prototyping, 3D printing, materials selection

Analysis and test

FEA, CFD, test planning, verification and validation, control systems, system identification, data analysis

Shop

Manual mill and lathe, laser marking, welding

Software

SolidWorks, Fusion 360, ANSYS (Mechanical, Fluent), MATLAB, Simulink, Python, Arduino, SQL, Power BI

Process

Lean Six Sigma (Green Belt trained), root cause analysis, CAPA